



SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



LOCATION: TRAIN 4	AREA: E
PERMIT TO WORK No.: —	TYPE OF SCAFFOLD: A Frame
SCAFFOLD DRAWING No: PMS-WA-ARCO-002	
RISK CLASSIFICATION: ☐ High ☒ Medium ☐ Low	

REFERENCE STANDARDS

- BS EN 1991-1-4:2005 — Wind Actions
- BS EN 1993-1-1:2005 — Design of Steel Structures (Eurocode 3)
- BS EN 12811-1:2003 — Temporary Works Equipment — Scaffolds

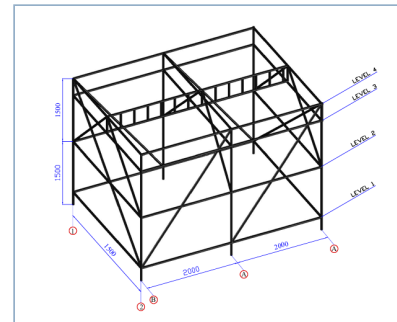
SCAFFOLD TO BE INSPECTED & TAGGED BEFORE USE

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of an **A Frame** scaffold required for *valve lifting* with equipment tag **412LICA-771** and work order **4258963**. The scaffold system consists of a **4.0m × 1.8m × 4.0m** A-Frame scaffold with vertical bracing for sway resistance. The structure is 15m above ground level.

SAFE WORKING LOAD (SWL): 3.125 kN (319 kg)

SITE / EQUIPMENT PHOTO



TOTAL HORIZONTAL LOAD SUMMARY

Direction	Wind (kN)	Impact (kN)	TOTAL (kN)
X	1.38	0.9375	2.317
Z	1.7	0.9375	2.638

Role	ID	Name	Signature	Date
DESIGNED BY	PMS/C3-1141	Nnamani, Ifeanyi		
VERIFIED BY	PMS/612C	Olowo, Peter		
REVIEWED BY	PIC/33C	Ezewoko, Joseph		
APPROVED BY	PIC/2	Chima, Tasieobi		

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	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	A Frame	Calculation Sheet	
	Document Number:	PMS-WA-ARCO-002	Prepared By: Nnamani, Ifeanyi	

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	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	A Frame	Calculation Sheet	
	Document Number:	PMS-WA-ARCO-002	Prepared By: Nnamani, Ifeanyi	

GENERAL NOTE

BS EN 12811-1, 4.2.1.2 Scaffold materials are in accordance with BS EN 12811 with minimum design strength of 210 N/mm². This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 – Design of Steel Structures).

Material Properties

48.3 mm outside diameter × 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 4.2.1.2 A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ($K_{FX}/K_{FZ} = 7.192$ kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot. See [Appendix A](#) for model views.

LOAD CASES

Load diagrams are shown in [Appendix B](#).

Dead Load / Self-weight Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Chain Hoist Load

BS EN 1991, 6.3.4.1 The equipment static weight is **255 kg ≈ 2.5 kN**. A **25% dynamic allowance** was applied for chain hoist operation.

Static weight = $255 \text{ kg} \times 9.81 / 1000 = \mathbf{2.5 \text{ kN}}$
Chain Hoist Design Load (CL) = $2.5 \times (1 + 0.25) = \mathbf{3.125 \text{ kN}}$

Impact Load

BS EN 1991, 6.3.4.1 30% of the chain hoist design load was assumed as a horizontal impact load, applied at the hoist attachment point independently in X and Z directions.

Impact Load (ILX) = $3.125 \times 0.30 = \mathbf{0.9375 \text{ kN}}$ (X-axis)
Impact Load (ILZ) = $3.125 \times 0.30 = \mathbf{0.9375 \text{ kN}}$ (Z-axis)

Wind Load

EN 1991-1-4:2005 Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.05678 kN/m** applied as a member UDL on all vertical standard members.

Total Wind Load — X Axis (WLX) = **1.38 kN**
Total Wind Load — Z Axis (WLZ) = **1.7 kN**

WIND LOAD CALCULATION — EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	$V_3 =$	36.0	m/s
	<i>Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity</i>		
Dredger P. — How to Approximate	Convert 3-second gust to 10-minute mean wind velocity		
	$V_3 / V_{3600} =$	1.52	
	$V_{600} / V_{3600} =$	1.06	
	$V_{600} = (V_{600}/V_{3600}) \times (V_{3600}/V_3) \times V_3 =$	25.11	m/s
	Basic wind speed $V_b = V_{600} =$	25.11	m/s
EN 1991-1-4:2005	Roughness Length (z_0) — Terrain Category 0	0.003	m
	Terrain category II ($z_{0,ii}$)	0.05	
EN 1991-1-4, Table 4.1	z_{min}	1.0	m
	z — height of structure under consideration Source: STAAD command geometry height	4.0	m
EN 1991-1-4, Sec 4.3	$C_{r0}(z)$ — Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	$C_o(z)$ — Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor $K_r = 0.19 \times [z_0/z_{0,ii}]^{0.07}$	0.156	
EN 1991-1-4, Table 4.1	$z_{min} < z \rightarrow$ Case 1 applies		
	Wind Turbulence		
	$C_r(z) = K_r \times \ln(z/z_0)$	1.1227	
	Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$	28.187	m/s
EN 1991-1-4, Sec 4.4	K_i	1.0	
	Turbulence intensity $I_v(z) = k_i / [C_o(z) \times \ln(z/z_0)]$	0.139	
	Peak Velocity Pressure		
	Density of air ρ	1.25	kg/m ³
EN 1991-1-4, Sec 4.5	$q_p(z) = [1 + 7 \cdot I_v(z)] \times 0.5 \times \rho \times V_m(z)^2$	979.6342	N/m ²
Wind Load = $C_f \times q_p \times A_f$			
$C_f =$		1.2	
$q_p =$		0.979634	kN/m ²
$A_f =$	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		0.05678	kN/m

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use $\gamma = 1.5$; SLS (characteristic) combinations use $\gamma = 1.0$.

DL = Dead Load | **CL** = Chain Hoist | **ILX/ILZ** = Impact X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	1.5DL + 1.5CL	$LC1 \times 1.5 + LC2 \times 1.5$
8	1.5DL + 1.5CL + 1.5ILX	$LC1 \times 1.5 + LC2 \times 1.5 + LC3 \times 1.5$
9	1.5DL + 1.5CL + 1.5ILZ	$LC1 \times 1.5 + LC2 \times 1.5 + LC4 \times 1.5$
10	1.5DL + 1.5WLX	$LC1 \times 1.5 + LC5 \times 1.5$
11	1.5DL + 1.5WLZ	$LC1 \times 1.5 + LC6 \times 1.5$

LC No.	Combination	Component Loadings & Factors
12	1.5DL + 1.5CL + 1.5WLX	$LC1 \times 1.5 + LC2 \times 1.5 + LC5 \times 1.5$
13	1.5DL + 1.5CL + 1.5WLZ	$LC1 \times 1.5 + LC2 \times 1.5 + LC6 \times 1.5$
14	DL + CL	$LC1 \times 1.0 + LC2 \times 1.0$
15	DL + CL + ILX	$LC1 \times 1.0 + LC2 \times 1.0 + LC3 \times 1.0$
16	DL + CL + ILZ	$LC1 \times 1.0 + LC2 \times 1.0 + LC4 \times 1.0$
17	DL + WLX	$LC1 \times 1.0 + LC5 \times 1.0$
18	DL + WLZ	$LC1 \times 1.0 + LC6 \times 1.0$
19	DL + CL + WLX	$LC1 \times 1.0 + LC2 \times 1.0 + LC5 \times 1.0$
20	DL + CL + WLZ	$LC1 \times 1.0 + LC2 \times 1.0 + LC6 \times 1.0$

STATIC CHECK PARAMETERS — UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.355 at Member No. 49, Load Combination 9, Clause EC-6.3.3-662. **PASS** — All members adequate; UC ratio below unity.

UC ratio diagram: see [Appendix C](#).

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
49	PIP483.6	PASS	EC-6.3.3-662	0.355	9	8.40	C
52	PIP483.6	PASS	EC-6.3.3-662	0.355	9	8.40	C
59	PIP483.6	PASS	EC-6.2.9.1	0.352	8	0.01	C
47	PIP483.6	PASS	EC-6.2.9.1	0.343	12	0.01	C
56	PIP483.6	PASS	EC-6.2.9.1	0.326	8	0.04	C
50	PIP483.6	PASS	EC-6.2.9.1	0.320	12	0.04	C
28	PIP483.6	PASS	EC-6.3.3-662	0.313	9	2.33	C
37	PIP483.6	PASS	EC-6.3.3-662	0.313	9	2.33	C
36	PIP483.6	PASS	EC-6.2.9.1	0.300	9	2.21	T
39	PIP483.6	PASS	EC-6.2.9.1	0.300	9	2.21	T
26	PIP483.6	PASS	EC-6.3.3-662	0.298	8	2.06	C
33	PIP483.6	PASS	EC-6.2.9.1	0.289	8	2.11	T
38	PIP483.6	PASS	EC-6.3.3-662	0.286	8	2.09	C
51	PIP483.6	PASS	EC-6.2.9.1	0.284	9	8.40	T
54	PIP483.6	PASS	EC-6.2.9.1	0.284	9	8.40	T
58	PIP483.6	PASS	EC-6.2.9.1	0.282	9	1.85	C
41	PIP483.6	PASS	EC-6.2.9.1	0.281	9	1.85	C
40	PIP483.6	PASS	EC-6.2.9.1	0.272	12	1.99	T
32	PIP483.6	PASS	EC-6.2.9.1	0.244	9	1.14	C
34	PIP483.6	PASS	EC-6.2.9.1	0.244	9	1.23	C
43	PIP483.6	PASS	EC-6.2.9.1	0.244	9	1.85	T
45	PIP483.6	PASS	EC-6.2.9.1	0.244	9	1.84	T
29	PIP483.6	PASS	EC-6.2.9.1	0.229	12	1.10	C
30	PIP483.6	PASS	EC-6.2.9.1	0.226	8	2.91	C
46	PIP483.6	PASS	EC-6.3.3-662	0.221	9	5.26	C
55	PIP483.6	PASS	EC-6.3.3-662	0.220	9	5.26	C
48	PIP483.6	PASS	EC-6.2.9.1	0.208	9	5.26	T
57	PIP483.6	PASS	EC-6.2.9.1	0.207	9	5.26	T
42	PIP483.6	PASS	EC-6.2.9.1	0.203	8	0.05	C
44	PIP483.6	PASS	EC-6.2.9.1	0.188	9	0.06	C
8	PIP483.6	PASS	EC-6.3.3-661	0.159	13	4.40	C

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
65	PIP483.6	PASS	EC-6.3.3-661	0.141	8	2.07	C
7	PIP483.6	PASS	EC-6.3.3-661	0.140	11	2.47	C
6	PIP483.6	PASS	EC-6.3.3-662	0.129	8	4.52	C
16	PIP483.6	PASS	EC-6.3.3-662	0.112	9	4.80	C
15	PIP483.6	PASS	EC-6.3.3-662	0.111	9	4.73	C
5	PIP483.6	PASS	EC-6.3.3-662	0.100	8	2.50	C
14	PIP483.6	PASS	EC-6.3.3-662	0.098	9	1.65	C
22	PIP483.6	PASS	EC-6.3.3-662	0.097	8	4.10	C
13	PIP483.6	PASS	EC-6.3.3-662	0.096	9	1.57	C
21	PIP483.6	PASS	EC-6.3.3-662	0.086	12	2.35	C
9	PIP483.6	PASS	EC-6.3.3-662	0.083	12	0.36	C
10	PIP483.6	PASS	EC-6.3.3-662	0.082	11	0.55	C
12	PIP483.6	PASS	EC-6.3.3-662	0.082	11	0.55	C
23	PIP483.6	PASS	EC-6.3.3-662	0.075	12	2.03	C
53	PIP483.6	PASS	EC-6.3.3-661	0.075	9	3.62	C
24	PIP483.6	PASS	EC-6.3.3-662	0.073	13	2.30	C
25	PIP483.6	PASS	EC-6.3.3-662	0.066	8	0.73	C
11	PIP483.6	PASS	EC-6.2.9.1	0.060	10	0.39	T
27	PIP483.6	PASS	EC-6.2.9.1	0.055	8	0.26	T
35	PIP483.6	PASS	EC-6.2.9.1	0.046	8	0.28	T
31	PIP483.6	PASS	EC-6.2.9.1	0.045	10	0.04	T
60	PIP483.6	PASS	EC-6.3.3-662	0.036	9	1.36	C
62	PIP483.6	PASS	EC-6.3.3-662	0.036	9	1.37	C
66	PIP483.6	PASS	EC-6.2.9.1	0.033	7	0.02	C
19	PIP483.6	PASS	EC-6.3.3-662	0.029	8	0.10	C
17	PIP483.6	PASS	EC-6.2.9.1	0.023	10	0.07	T
61	PIP483.6	PASS	EC-6.2.9.1	0.019	7	0.01	C
63	PIP483.6	PASS	EC-6.2.9.1	0.019	7	0.01	T
18	PIP483.6	PASS	EC-6.2.9.1	0.014	9	0.30	T
20	PIP483.6	PASS	EC-6.2.9.1	0.014	9	0.30	T

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **6.290 kN** (Member 52, C). **Right-angle double coupler of Class A (10 kN slipping resistance)** shall be used for all scaffold steel tube connections at node joints.

Check: 6.290 kN < 10.0 kN → **PASS**

Member force table: see [Appendix D](#).

Table C.1 — Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force F _{s,k} (kN)	10.0	15.0
	Cruciform bending moment M _{B,k} (kNm)	—	0.8
	Pull-apart force F _{p,k} (kN)	20.0	30.0
	Rotational moment M _{T,k} (kNm)	—	0.13

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Swivel Coupler (SW)	Slipping force $F_{s,k}$ (kN)	6.0	9.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.8
	Pull-apart force $F_{p,k}$ (kN)	15.0	22.5
	Rotational moment $M_{T,k}$ (kNm)	—	0.08
Putlog / Single Coupler (PL)	Slipping force $F_{s,k}$ (kN)	—	3.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.6

FRICTIONAL RESISTANCE

Max vertical reaction per support ($\text{Max } f_y$) = Resistance $\times 0.15 / C_F$	= 3.596 kN
Coefficient of friction (C_F)	= 0.3
Frictional Resistance (FR) = $C_F \times \text{Max } f_y$	= 1.0788 kN
Spring Stiffness = FR / 0.15	= 7.192 kN ← applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results. Diagrams and displacement tables are in [Appendix C](#) and [Appendix D](#).

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement — LC 15 Member 11 Member length L = 300.0 mm	0.492 mm	$L/100 = 300/100 = 3.0 \text{ mm}$	PASS

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement — LC 19 Member 16 Member length L = 1800.0 mm	0.307 mm	$L/200 = 1800/200 = 9.0 \text{ mm}$	PASS

CONCLUSION

The structural analysis and design of the **A Frame** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.355** at Member 49). Vertical and horizontal deflections are within their respective allowable limits of **L/100 = 3.0mm** and **L/200 = 9.0mm**. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and material specification. Any proposed change to the design requires a re-design and written approval from the Scaffold Engineer.

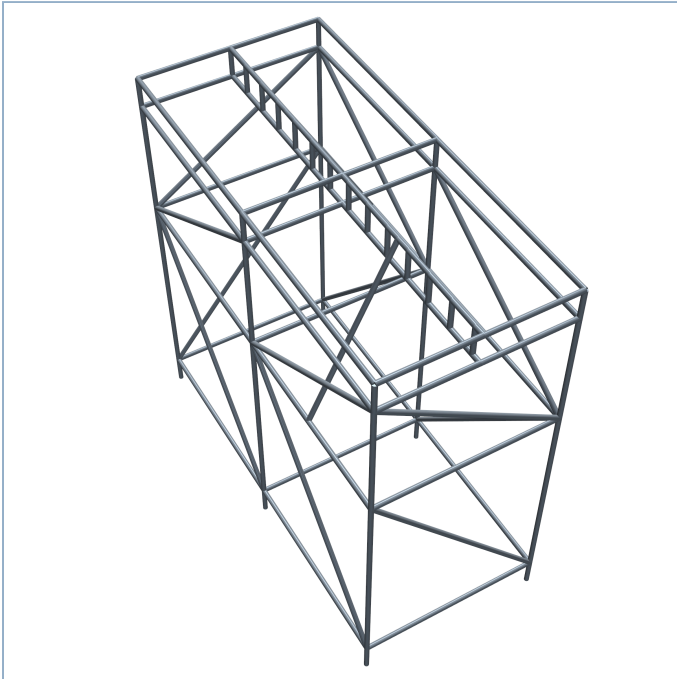
INSTALLATION NOTES

1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.

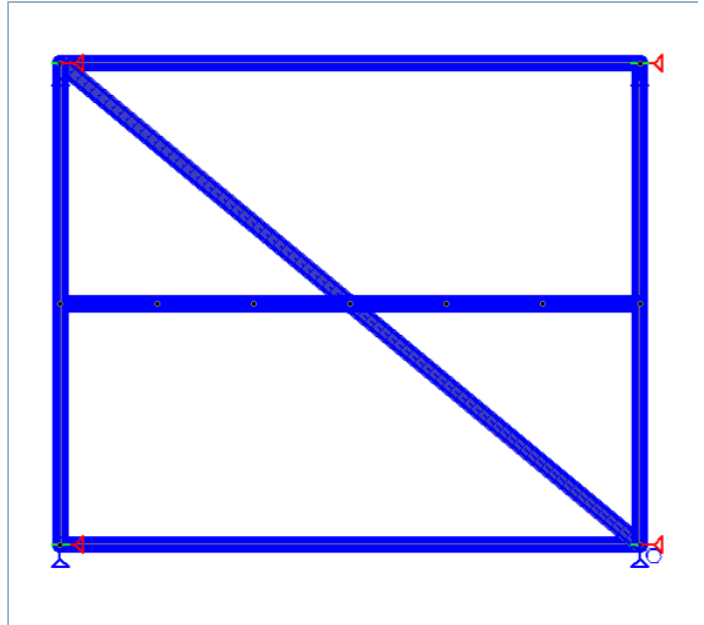
2. The work vicinity must be barricaded and appropriate signage placed at all access points.
3. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
4. A valid work permit must be obtained, discussed, and signed before erection commences.
5. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
6. Lay out and level the four base plates at the standard positions, ensuring a plan footprint of 4.0m x 1.8m in accordance with the approved working drawing.
7. Erect the four scaffold tube standards (48.3 x 3.6 mm) vertically on the base plates and secure with base collars. Check plumb in both axes before proceeding.
8. Install the lower horizontal ledgers and transoms at the kicker lift height (0.300m) to tie the four standards together and form the rigid base frame.
9. Install mid-level horizontal ledgers and transoms at 2.5m height in accordance with the working drawing.
10. Install mid-level horizontal ledgers and transoms at 3.7m height in accordance with the working drawing.
11. Fix all vertical bracing members diagonally across the face panels of the A-frame structure using right-angle couplers to provide sway resistance in both X and Z axes as shown on the working drawing.
12. Install the 4000mm ladder beam horizontally across the top of the standards at 4.0m height, connecting to both standards with right-angle double couplers. Ensure the beam is level before fully tightening all couplers.
13. Rig the chain hoist to the ladder beam at the designated hoist point shown on the working drawing. Confirm the hoist is rated for a minimum safe working load equal to or greater than the design SWL.
14. The chain hoist attachment point and all coupler connections at ladder beam level shall be inspected and torque-checked by a competent person before any lifting operation commences.
15. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before any lift is performed.
16. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
17. All modifications made after erection must be carried out by a competent scaffold team with the written approval of the scaffold engineer. End-user or third-party modification of this scaffold structure is strictly prohibited.
18. For dismantling, perform the erection steps in reverse order.

APPENDIX A — MODEL VIEWS

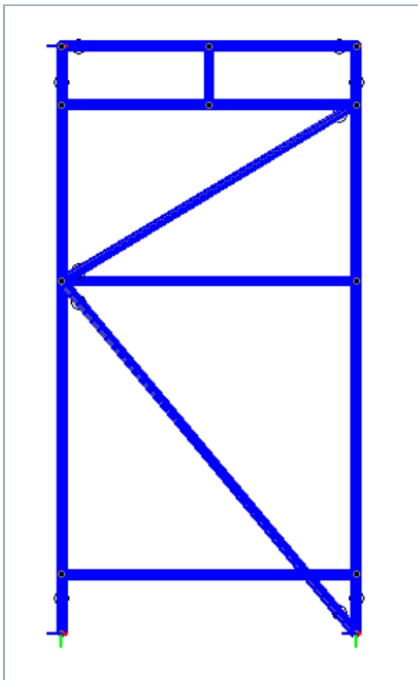
3D Isometric View



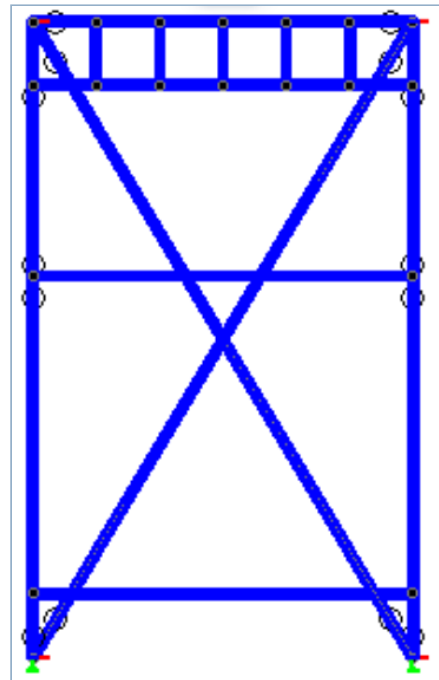
Plan View



Elevation from +X Axis



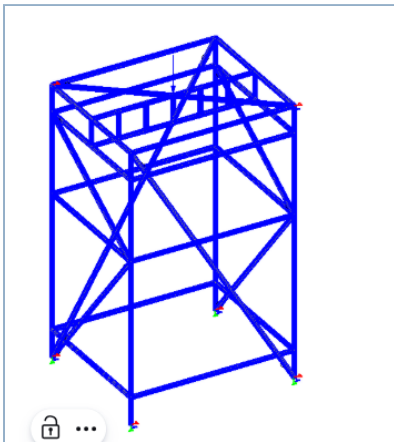
Elevation from +Z Axis



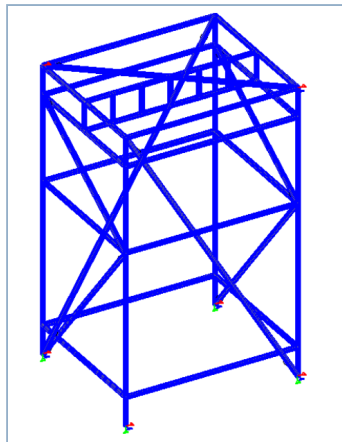
Scaffold Dimensions: W = 4.0m | D = 1.8m | H = 4.0m | **Lifts:** 0.3m, 2.5m, 3.7m, 4.0m

APPENDIX B — LOAD DIAGRAMS

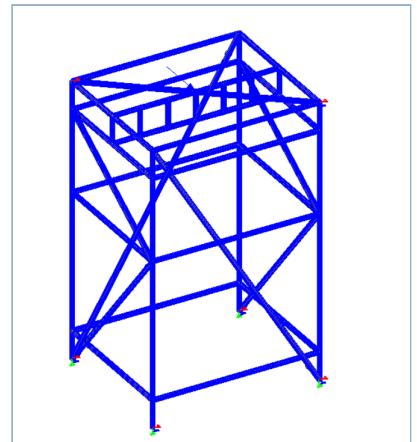
Chain Hoist Load (CL)



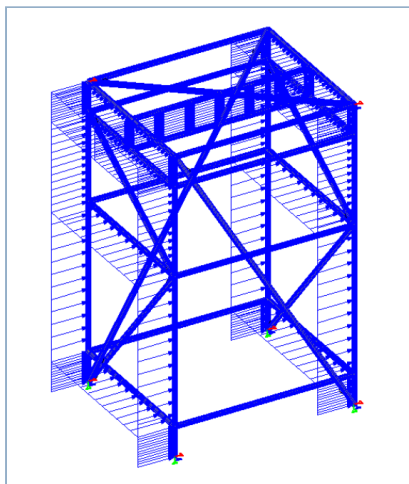
Impact Load — X Axis (ILX)



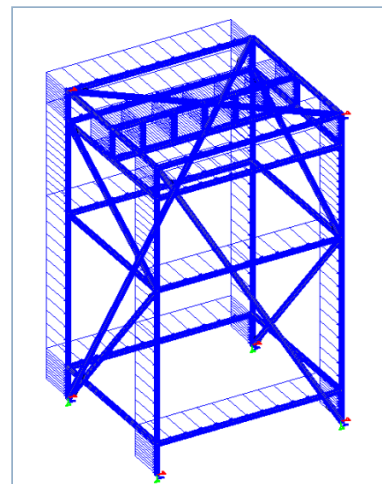
Impact Load — Z Axis (ILZ)



Wind Load — X Axis (WLX)

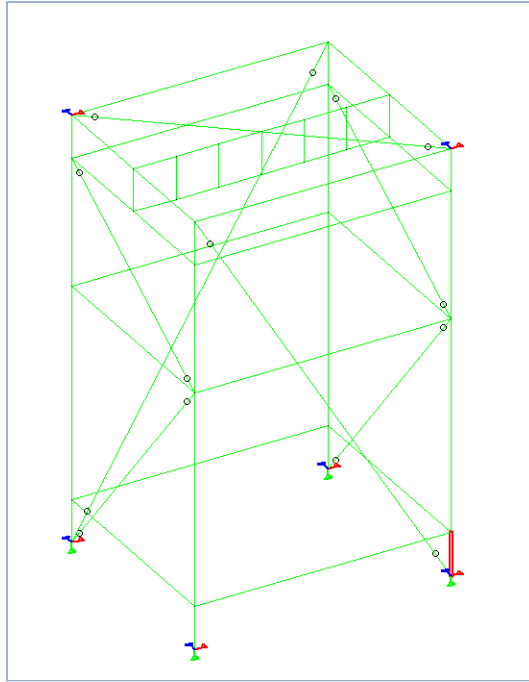


Wind Load — Z Axis (WLZ)

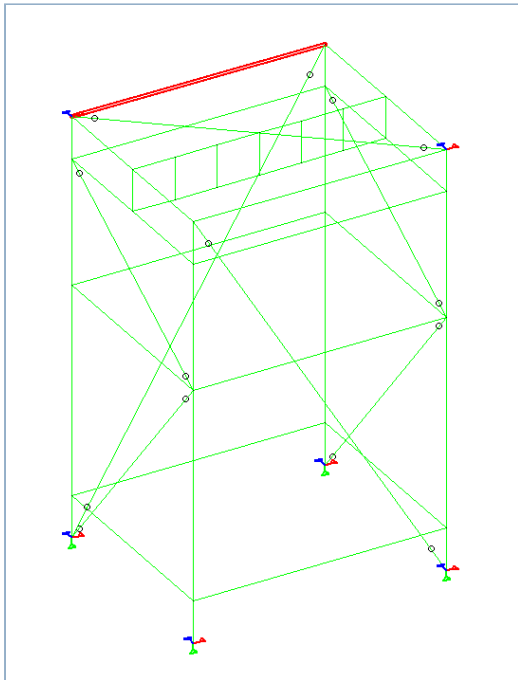


APPENDIX C — ANALYSIS RESULTS

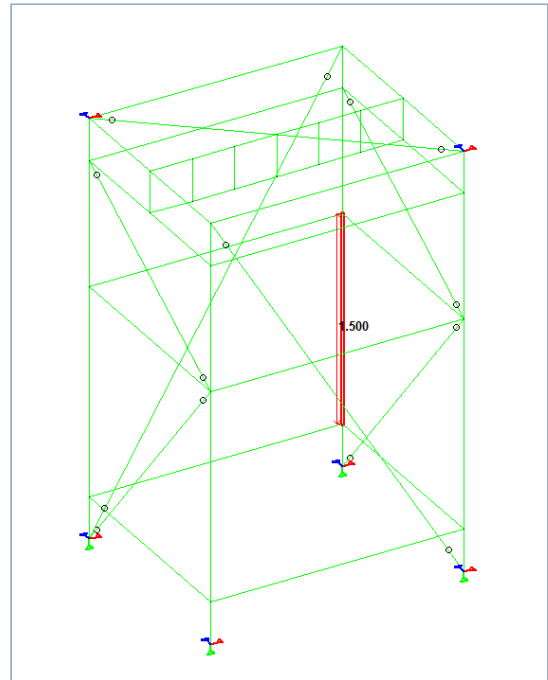
Utilization Ratio (UC) Diagram



Vertical Deflection Diagram



Horizontal Deflection Diagram



APPENDIX D — GUI RESULTS TABLES

Member Force / Code Check Table (Connection Stability)

Beam	Analysis Property	Design Property	Actual Ratio	Allowable Ratio	Normalized Ratio (Actual/Allowable)	Clause	L/C	Ax m2	r
59	PIP483.6	PIP483.6	0.796	1.000	0.796	EC-6.3.3-662	7	0.001	0.000
128	PIP483.6	PIP483.6	0.796	1.000	0.796	EC-6.3.3-662	7	0.001	0.000
48	PIP483.6	PIP483.6	0.598	1.000	0.598	EC-6.2.9.1	7	0.001	0.000
51	PIP483.6	PIP483.6	0.593	1.000	0.593	EC-6.2.9.1	7	0.001	0.000
55	PIP483.6	PIP483.6	0.389	1.000	0.389	EC-6.2.9.1	10	0.001	0.000
60	PIP483.6	PIP483.6	0.384	1.000	0.384	EC-6.2.9.1	10	0.001	0.000
90	PIP483.6	PIP483.6	0.372	1.000	0.372	EC-6.2.9.1	11	0.001	0.000
124	PIP483.6	PIP483.6	0.372	1.000	0.372	EC-6.2.9.1	11	0.001	0.000
93	PIP483.6	PIP483.6	0.362	1.000	0.362	EC-6.2.9.1	11	0.001	0.000
91	PIP483.6	PIP483.6	0.354	1.000	0.354	EC-6.2.9.1	11	0.001	0.000
88	PIP483.6	PIP483.6	0.345	1.000	0.345	EC-6.2.9.1	11	0.001	0.000
87	PIP483.6	PIP483.6	0.341	1.000	0.341	EC-6.2.9.1	11	0.001	0.000
94	PIP483.6	PIP483.6	0.334	1.000	0.334	EC-6.2.9.1	11	0.001	0.000
96	PIP483.6	PIP483.6	0.332	1.000	0.332	EC-6.2.9.1	11	0.001	0.000
42	PIP483.6	PIP483.6	0.323	1.000	0.323	EC-6.3.3-662	10	0.001	0.000
85	PIP483.6	PIP483.6	0.311	1.000	0.311	EC-6.2.9.1	11	0.001	0.000
97	PIP483.6	PIP483.6	0.309	1.000	0.309	EC-6.2.9.1	11	0.001	0.000
61	PIP483.6	PIP483.6	0.306	1.000	0.306	EC-6.3.3-662	10	0.001	0.000
84	PIP483.6	PIP483.6	0.302	1.000	0.302	EC-6.2.9.1	11	0.001	0.000
99	PIP483.6	PIP483.6	0.297	1.000	0.297	EC-6.2.9.1	11	0.001	0.000
100	PIP483.6	PIP483.6	0.274	1.000	0.274	EC-6.2.9.1	11	0.001	0.000
82	PIP483.6	PIP483.6	0.268	1.000	0.268	EC-6.2.9.1	11	0.001	0.000
81	PIP483.6	PIP483.6	0.259	1.000	0.259	EC-6.2.9.1	11	0.001	0.000
102	PIP483.6	PIP483.6	0.255	1.000	0.255	EC-6.2.9.1	11	0.001	0.000
20	PIP483.6	PIP483.6	0.243	1.000	0.243	EC-6.2.9.1	11	0.001	0.000
103	PIP483.6	PIP483.6	0.232	1.000	0.232	EC-6.2.9.1	11	0.001	0.000
30	PIP483.6	PIP483.6	0.229	1.000	0.229	EC-6.2.9.1	10	0.001	0.000
120	PIP483.6	PIP483.6	0.229	1.000	0.229	EC-6.2.9.1	10	0.001	0.000
16	PIP483.6	PIP483.6	0.221	1.000	0.221	EC-6.3.3-662	11	0.001	0.000
79	PIP483.6	PIP483.6	0.216	1.000	0.216	EC-6.2.9.1	11	0.001	0.000
11	PIP483.6	PIP483.6	0.213	1.000	0.213	EC-6.3.3-662	11	0.001	0.000
10	PIP483.6	PIP483.6	0.208	1.000	0.208	EC-6.2.9.1	11	0.001	0.000
78	PIP483.6	PIP483.6	0.208	1.000	0.208	EC-6.2.9.1	11	0.001	0.000
105	PIP483.6	PIP483.6	0.205	1.000	0.205	EC-6.2.9.1	11	0.001	0.000
17	PIP483.6	PIP483.6	0.196	1.000	0.196	EC-6.2.9.1	11	0.001	0.000
121	PIP483.6	PIP483.6	0.194	1.000	0.194	EC-6.2.9.1	11	0.001	0.000
15	PIP483.6	PIP483.6	0.192	1.000	0.192	EC-6.2.9.1	11	0.001	0.000
27	PIP483.6	PIP483.6	0.190	1.000	0.190	EC-6.2.9.1	11	0.001	0.000
64	PIP483.6	PIP483.6	0.182	1.000	0.182	EC-6.2.9.1	11	0.001	0.000
106	PIP483.6	PIP483.6	0.182	1.000	0.182	EC-6.2.9.1	11	0.001	0.000

Vertical Displacement Table (SLS Combinations)

Beam	Analysis Property	Design Property	Actual Ratio	Allowable Ratio	Normalized Ratio (Actual/Allowable)	Clause	L/C	Ax m2	Δ
59	PIP483.6	PIP483.6	0.796	1.000	0.796	EC-6.3.3-662	7	0.001	0.000
128	PIP483.6	PIP483.6	0.796	1.000	0.796	EC-6.3.3-662	7	0.001	0.000
48	PIP483.6	PIP483.6	0.598	1.000	0.598	EC-6.2.9.1	7	0.001	0.000
51	PIP483.6	PIP483.6	0.593	1.000	0.593	EC-6.2.9.1	7	0.001	0.000
55	PIP483.6	PIP483.6	0.389	1.000	0.389	EC-6.2.9.1	10	0.001	0.000
60	PIP483.6	PIP483.6	0.384	1.000	0.384	EC-6.2.9.1	10	0.001	0.000
90	PIP483.6	PIP483.6	0.372	1.000	0.372	EC-6.2.9.1	11	0.001	0.000
124	PIP483.6	PIP483.6	0.372	1.000	0.372	EC-6.2.9.1	11	0.001	0.000
93	PIP483.6	PIP483.6	0.362	1.000	0.362	EC-6.2.9.1	11	0.001	0.000
91	PIP483.6	PIP483.6	0.354	1.000	0.354	EC-6.2.9.1	11	0.001	0.000
88	PIP483.6	PIP483.6	0.345	1.000	0.345	EC-6.2.9.1	11	0.001	0.000
87	PIP483.6	PIP483.6	0.341	1.000	0.341	EC-6.2.9.1	11	0.001	0.000
94	PIP483.6	PIP483.6	0.334	1.000	0.334	EC-6.2.9.1	11	0.001	0.000
96	PIP483.6	PIP483.6	0.332	1.000	0.332	EC-6.2.9.1	11	0.001	0.000
42	PIP483.6	PIP483.6	0.323	1.000	0.323	EC-6.3.3-662	10	0.001	0.000
85	PIP483.6	PIP483.6	0.311	1.000	0.311	EC-6.2.9.1	11	0.001	0.000
97	PIP483.6	PIP483.6	0.309	1.000	0.309	EC-6.2.9.1	11	0.001	0.000
61	PIP483.6	PIP483.6	0.306	1.000	0.306	EC-6.3.3-662	10	0.001	0.000
84	PIP483.6	PIP483.6	0.302	1.000	0.302	EC-6.2.9.1	11	0.001	0.000
99	PIP483.6	PIP483.6	0.297	1.000	0.297	EC-6.2.9.1	11	0.001	0.000
100	PIP483.6	PIP483.6	0.274	1.000	0.274	EC-6.2.9.1	11	0.001	0.000
82	PIP483.6	PIP483.6	0.268	1.000	0.268	EC-6.2.9.1	11	0.001	0.000
81	PIP483.6	PIP483.6	0.259	1.000	0.259	EC-6.2.9.1	11	0.001	0.000
102	PIP483.6	PIP483.6	0.255	1.000	0.255	EC-6.2.9.1	11	0.001	0.000
20	PIP483.6	PIP483.6	0.243	1.000	0.243	EC-6.2.9.1	11	0.001	0.000
103	PIP483.6	PIP483.6	0.232	1.000	0.232	EC-6.2.9.1	11	0.001	0.000
30	PIP483.6	PIP483.6	0.229	1.000	0.229	EC-6.2.9.1	10	0.001	0.000
120	PIP483.6	PIP483.6	0.229	1.000	0.229	EC-6.2.9.1	10	0.001	0.000
16	PIP483.6	PIP483.6	0.221	1.000	0.221	EC-6.3.3-662	11	0.001	0.000
79	PIP483.6	PIP483.6	0.216	1.000	0.216	EC-6.2.9.1	11	0.001	0.000
11	PIP483.6	PIP483.6	0.213	1.000	0.213	EC-6.3.3-662	11	0.001	0.000
10	PIP483.6	PIP483.6	0.208	1.000	0.208	EC-6.2.9.1	11	0.001	0.000
78	PIP483.6	PIP483.6	0.208	1.000	0.208	EC-6.2.9.1	11	0.001	0.000
105	PIP483.6	PIP483.6	0.205	1.000	0.205	EC-6.2.9.1	11	0.001	0.000
17	PIP483.6	PIP483.6	0.196	1.000	0.196	EC-6.2.9.1	11	0.001	0.000
121	PIP483.6	PIP483.6	0.194	1.000	0.194	EC-6.2.9.1	11	0.001	0.000
15	PIP483.6	PIP483.6	0.192	1.000	0.192	EC-6.2.9.1	11	0.001	0.000
27	PIP483.6	PIP483.6	0.190	1.000	0.190	EC-6.2.9.1	11	0.001	0.000
64	PIP483.6	PIP483.6	0.182	1.000	0.182	EC-6.2.9.1	11	0.001	0.000
106	PIP483.6	PIP483.6	0.182	1.000	0.182	EC-6.2.9.1	11	0.001	0.000

Max = 0.492 mm (LC 15) | L = Member 11 length (300 mm) | Allowable L/100 = 3.0 mm → **PASS**

Horizontal Displacement Table (SLS Combinations)

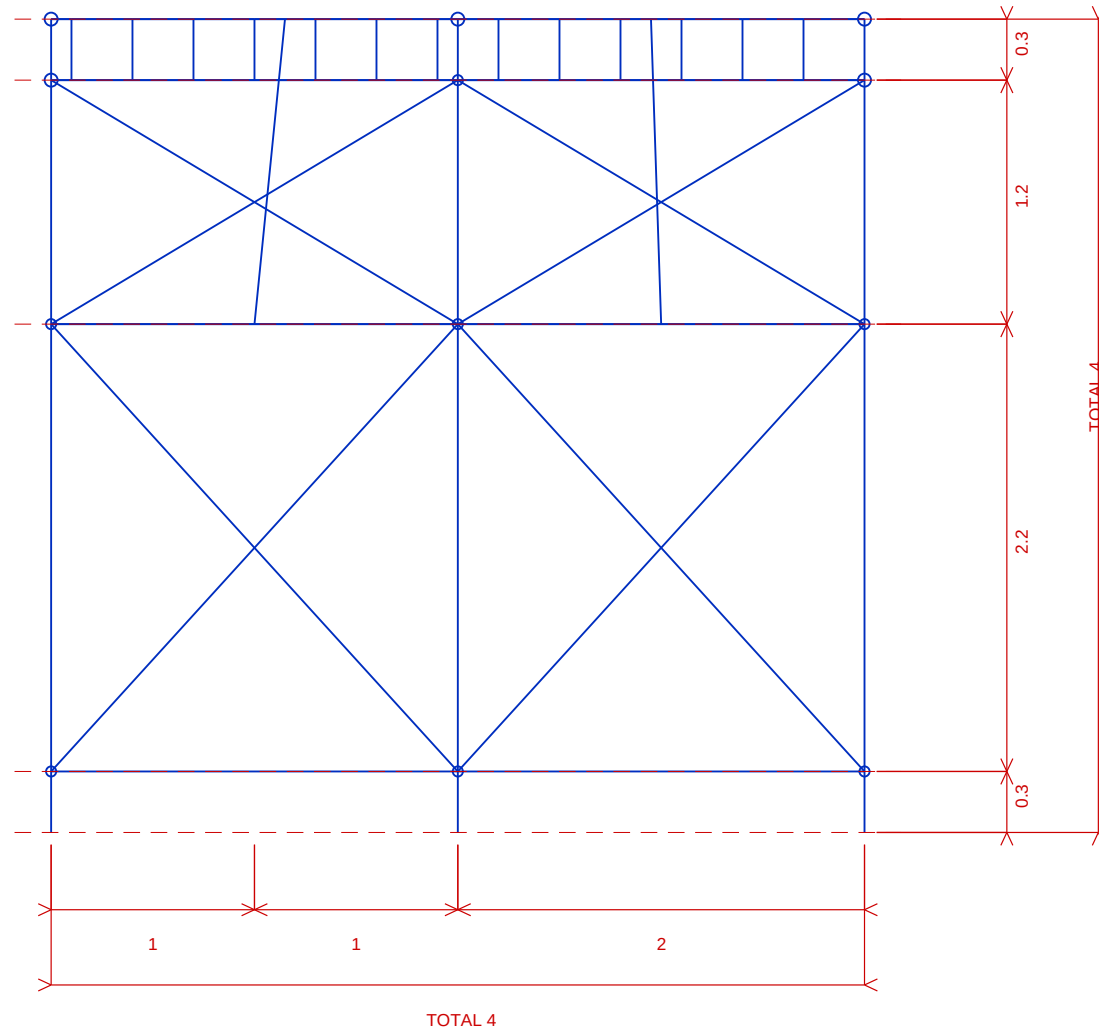
Beam	Analysis Property	Design Property	Actual Ratio	Allowable Ratio	Normalized Ratio (Actual/Allowable)	Clause	L/C	Ax m2	Limit
59	PIP483.6	PIP483.6	0.796	1.000	0.796	EC-6.3.3-662	7	0.001	0.000
128	PIP483.6	PIP483.6	0.796	1.000	0.796	EC-6.3.3-662	7	0.001	0.000
48	PIP483.6	PIP483.6	0.598	1.000	0.598	EC-6.2.9.1	7	0.001	0.000
51	PIP483.6	PIP483.6	0.593	1.000	0.593	EC-6.2.9.1	7	0.001	0.000
55	PIP483.6	PIP483.6	0.389	1.000	0.389	EC-6.2.9.1	10	0.001	0.000
60	PIP483.6	PIP483.6	0.384	1.000	0.384	EC-6.2.9.1	10	0.001	0.000
90	PIP483.6	PIP483.6	0.372	1.000	0.372	EC-6.2.9.1	11	0.001	0.000
124	PIP483.6	PIP483.6	0.372	1.000	0.372	EC-6.2.9.1	11	0.001	0.000
93	PIP483.6	PIP483.6	0.362	1.000	0.362	EC-6.2.9.1	11	0.001	0.000
91	PIP483.6	PIP483.6	0.354	1.000	0.354	EC-6.2.9.1	11	0.001	0.000
88	PIP483.6	PIP483.6	0.345	1.000	0.345	EC-6.2.9.1	11	0.001	0.000
87	PIP483.6	PIP483.6	0.341	1.000	0.341	EC-6.2.9.1	11	0.001	0.000
94	PIP483.6	PIP483.6	0.334	1.000	0.334	EC-6.2.9.1	11	0.001	0.000
96	PIP483.6	PIP483.6	0.332	1.000	0.332	EC-6.2.9.1	11	0.001	0.000
42	PIP483.6	PIP483.6	0.323	1.000	0.323	EC-6.3.3-662	10	0.001	0.000
85	PIP483.6	PIP483.6	0.311	1.000	0.311	EC-6.2.9.1	11	0.001	0.000
97	PIP483.6	PIP483.6	0.309	1.000	0.309	EC-6.2.9.1	11	0.001	0.000
61	PIP483.6	PIP483.6	0.306	1.000	0.306	EC-6.3.3-662	10	0.001	0.000
84	PIP483.6	PIP483.6	0.302	1.000	0.302	EC-6.2.9.1	11	0.001	0.000
99	PIP483.6	PIP483.6	0.297	1.000	0.297	EC-6.2.9.1	11	0.001	0.000
100	PIP483.6	PIP483.6	0.274	1.000	0.274	EC-6.2.9.1	11	0.001	0.000
82	PIP483.6	PIP483.6	0.268	1.000	0.268	EC-6.2.9.1	11	0.001	0.000
81	PIP483.6	PIP483.6	0.259	1.000	0.259	EC-6.2.9.1	11	0.001	0.000
102	PIP483.6	PIP483.6	0.255	1.000	0.255	EC-6.2.9.1	11	0.001	0.000
20	PIP483.6	PIP483.6	0.243	1.000	0.243	EC-6.2.9.1	11	0.001	0.000
103	PIP483.6	PIP483.6	0.232	1.000	0.232	EC-6.2.9.1	11	0.001	0.000
30	PIP483.6	PIP483.6	0.229	1.000	0.229	EC-6.2.9.1	10	0.001	0.000
120	PIP483.6	PIP483.6	0.229	1.000	0.229	EC-6.2.9.1	10	0.001	0.000
16	PIP483.6	PIP483.6	0.221	1.000	0.221	EC-6.3.3-662	11	0.001	0.000
79	PIP483.6	PIP483.6	0.216	1.000	0.216	EC-6.2.9.1	11	0.001	0.000
11	PIP483.6	PIP483.6	0.213	1.000	0.213	EC-6.3.3-662	11	0.001	0.000
10	PIP483.6	PIP483.6	0.208	1.000	0.208	EC-6.2.9.1	11	0.001	0.000
78	PIP483.6	PIP483.6	0.208	1.000	0.208	EC-6.2.9.1	11	0.001	0.000
105	PIP483.6	PIP483.6	0.205	1.000	0.205	EC-6.2.9.1	11	0.001	0.000
17	PIP483.6	PIP483.6	0.196	1.000	0.196	EC-6.2.9.1	11	0.001	0.000
121	PIP483.6	PIP483.6	0.194	1.000	0.194	EC-6.2.9.1	11	0.001	0.000
15	PIP483.6	PIP483.6	0.192	1.000	0.192	EC-6.2.9.1	11	0.001	0.000
27	PIP483.6	PIP483.6	0.190	1.000	0.190	EC-6.2.9.1	11	0.001	0.000
64	PIP483.6	PIP483.6	0.182	1.000	0.182	EC-6.2.9.1	11	0.001	0.000
106	PIP483.6	PIP483.6	0.182	1.000	0.182	EC-6.2.9.1	11	0.001	0.000

Max = 0.307 mm (LC 19) | L = Member 16 length (1800 mm) | Allowable L/200 = 9.0 mm → **PASS**

APPENDIX E — REFERENCES & STANDARDS

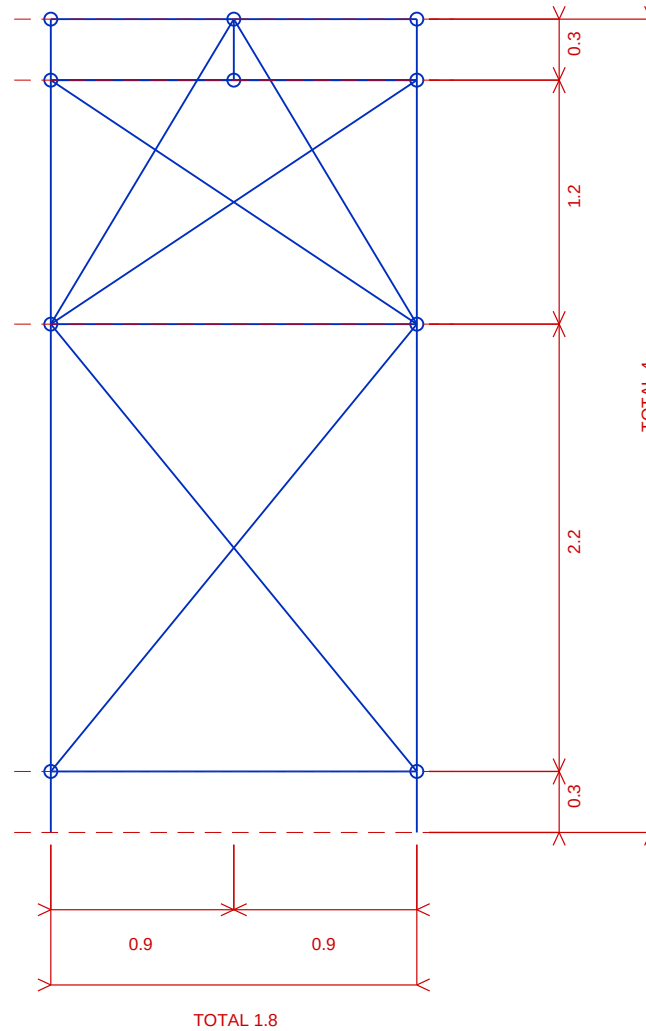
Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment — Scaffolds — Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures — General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures — Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Dredger P.	How to Approximate 10-Minute Mean Wind Speed from 3-Second Gust	—
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 — Structural Analysis and Design Software	2022

Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.
Report: A-Frame Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



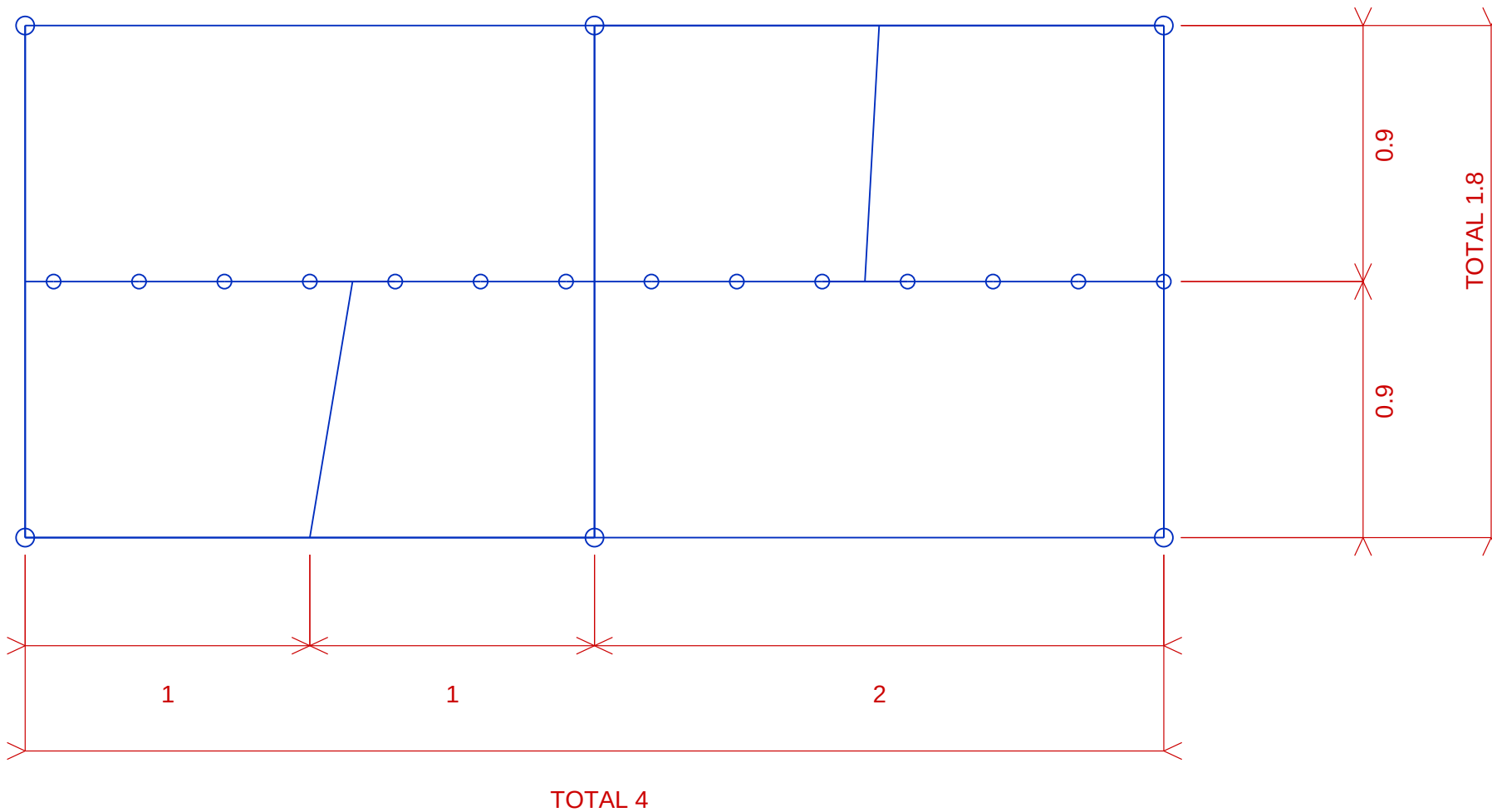
FRONT VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-FRONT-VIEW				
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - FRONT VIEW		SCALE NTS	DATE 05-JUN-2026	REV 0		
CLIENT 	LOCATION 	DRAWN Nnamani, Ifeanyi	CHECKED 	APPROVED 	COMPANY 	
					SHEET 1 OF 4	SIZE A4



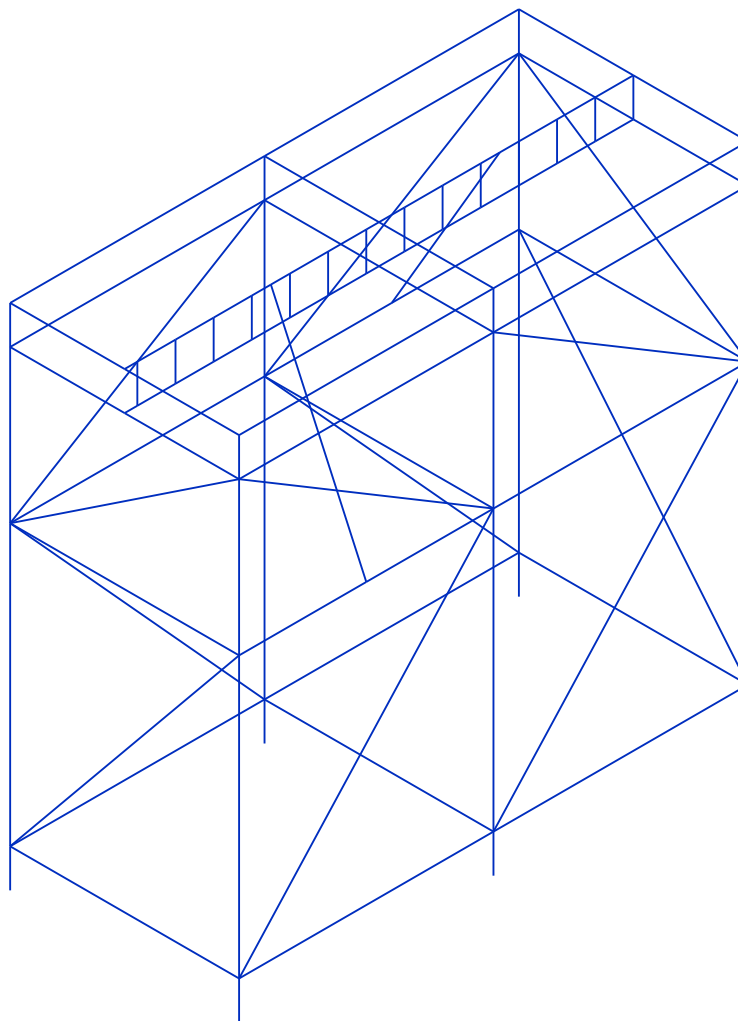
SIDE VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-SIDE-VIEW			 
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - SIDE VIEW		SCALE NTS	DATE 05-JUN-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY
					SHEET 2 OF 4 SIZE A4



PLAN VIEW

PROJECT		DRAWING NO.			 NLNG	
SCAFFOLDING WORKS		PMS-WA-ARCO-003-PLAN-VIEW				
DRAWING TITLE		SCALE	DATE	REV	COMPANY	
SCAFFOLD GENERAL ARRANGEMENT - PLAN VIEW		NTS	05-JUN-2026	0		
CLIENT	LOCATION	DRAWN	CHECKED	APPROVED	SHEET	
		Nnamani, Ifeanyi				
					3 OF 4	A4



3D VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-3D-VIEW			 NLNG	
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - 3D VIEW		SCALE NTS	DATE 05-JUN-2026	REV 0		
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY	
					SHEET 4 OF 4	SIZE A4

3D Isometric View

